TRAPEZOIDAL RULE

- Another approximation—called the Trapezoidal Rule—results from averaging the approximations in Equations 1 and 2, as follows.

$$\int_a^b f(x)dx \approx \frac{1}{2}\left[\sum_{i=1}^{n} f(x_{i-1})\Delta x + \sum_{i=1}^{n} f(x_i)\Delta x\right]$$

$$= \frac{\Delta x}{2}\left[\sum_{i=1}^{n}(f(x_{i-1}) + f(x_i))\right]$$

$$= \frac{\Delta x}{2}[(f(x_0) + f(x_1)) + (f(x_1) + f(x_2)) \quad +\ldots + (f(x_{n-1}) + f(x_n))]$$

$$= \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + 2f(x_2) \quad +\ldots + 2f(x_{n-1}) + f(x_n)]$$

TRAPEZOIDAL RULE

Approximate integration results from using straight line segments to approximate a curve.

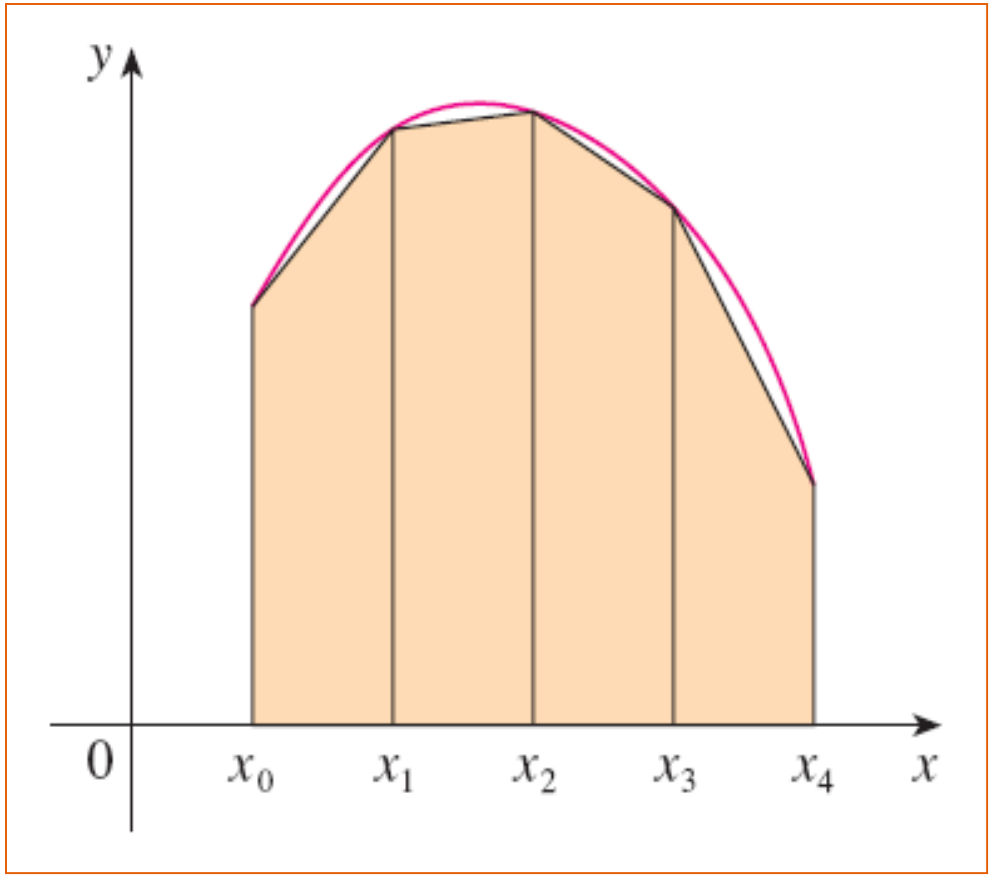
$$\int_a^b f(x)dx \approx T_n$$

$$= \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + 2f(x_2) \ + \ldots + 2f(x_{n-1}) + f(x_n)]$$

- where $\Delta x = (b - a)/n$ and $x_i = a + i\,\Delta x$

TRAPEZOIDAL RULE

- The reason for the name can be seen from the figure, which illustrates the case $f(x) \geq 0$.

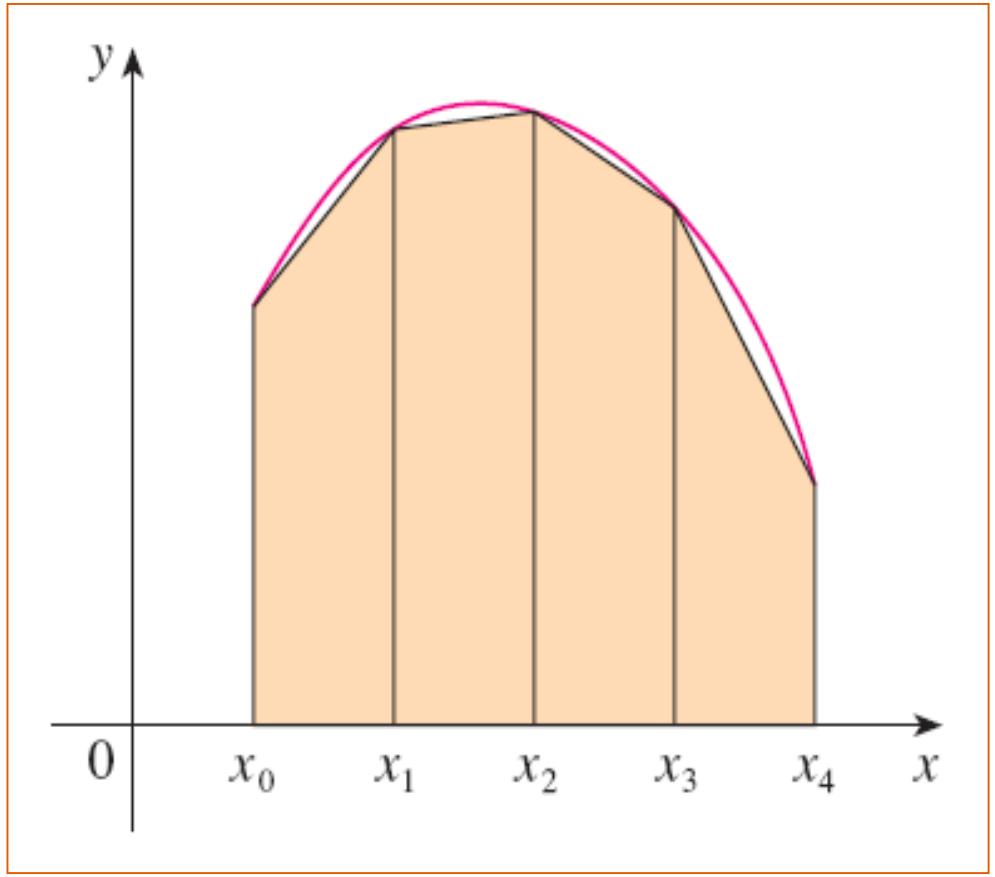


TRAPEZOIDAL RULE

- The area of the trapezoid that lies above the i_{th} subinterval is:

$$\Delta x\left(\frac{f(x_{i-1})+f(x_i)}{2}\right)=\frac{\Delta x}{2}[f(x_{i-1})+f(x_i)]$$

- If we add the areas of all these trapezoids, we get the right side of the Trapezoidal Rule.



APPROXIMATE INTEGRATION

Example 1(a)

- Approximate the integral with $n = 5$, using: $\int_1^2 (1/x)dx$
- With $n = 5$, $a = 1$ and $b = 2$, we have: $\Delta x = (2 - 1)/5 = 0.2$
  - **<u>So, the Trapezoidal Rule gives:</u>**

$$\int_1^2 \frac{1}{x}dx \approx T_5 = \frac{0.2}{2}[f(1) + 2f(1.2) + 2f(1.4) + 2f(1.6) + 2f(1.8) + f(2)]$$

$$= 0.1\left(\frac{1}{1} + \frac{2}{1.2} + \frac{2}{1.4} + \frac{2}{1.6} + \frac{2}{1.8} + \frac{1}{2}\right) \approx 0.695635$$

- $M_5 = 0.691908$

APPROXIMATE INTEGRATION

- Another rule for approximate integration results from using (quadratic or cubic) curves instead of straight line segments to approximate a curve.

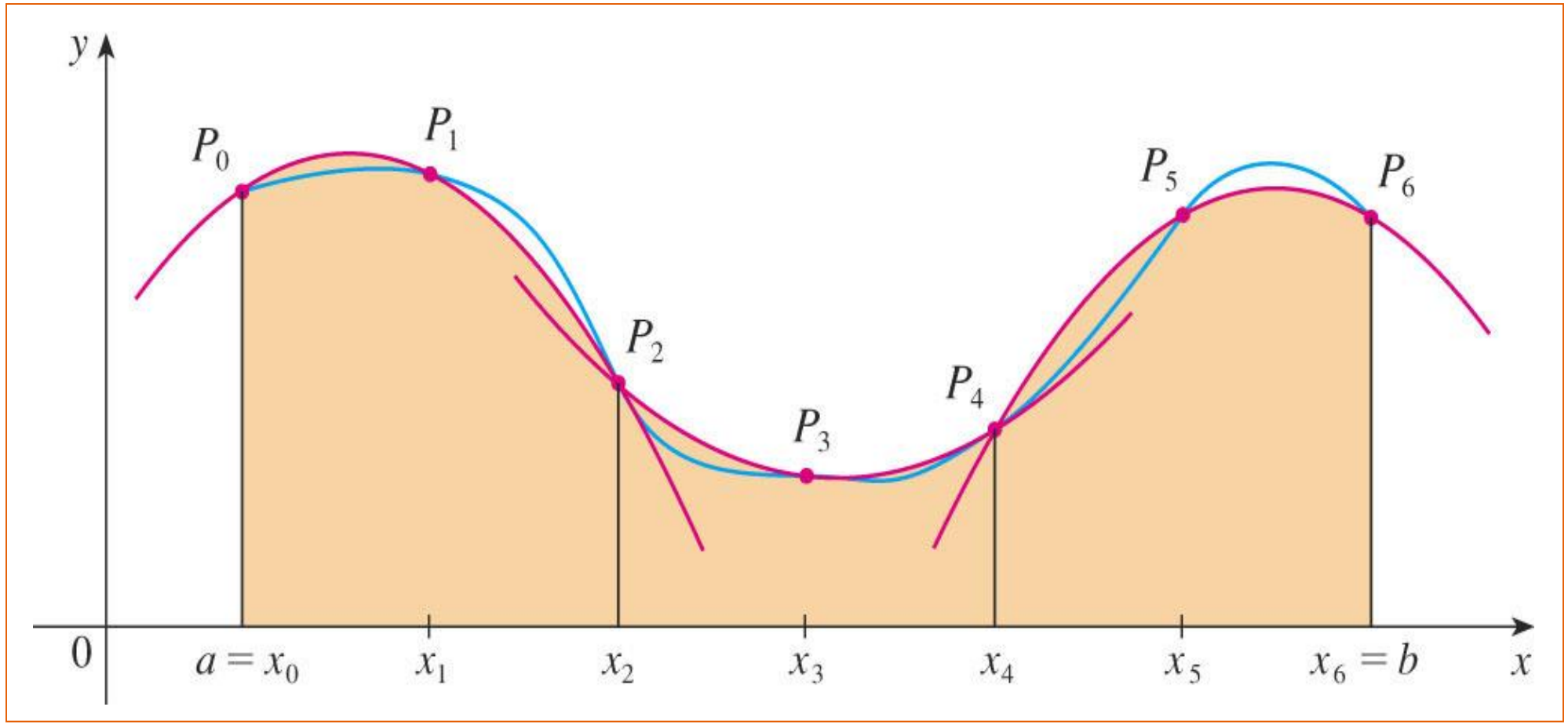
APPROXIMATE INTEGRATION

- As before, we divide $[a, b]$ into n subintervals of equal length $h = \Delta x = (b - a)/n$.

- However, this time, we assume n is an even number.

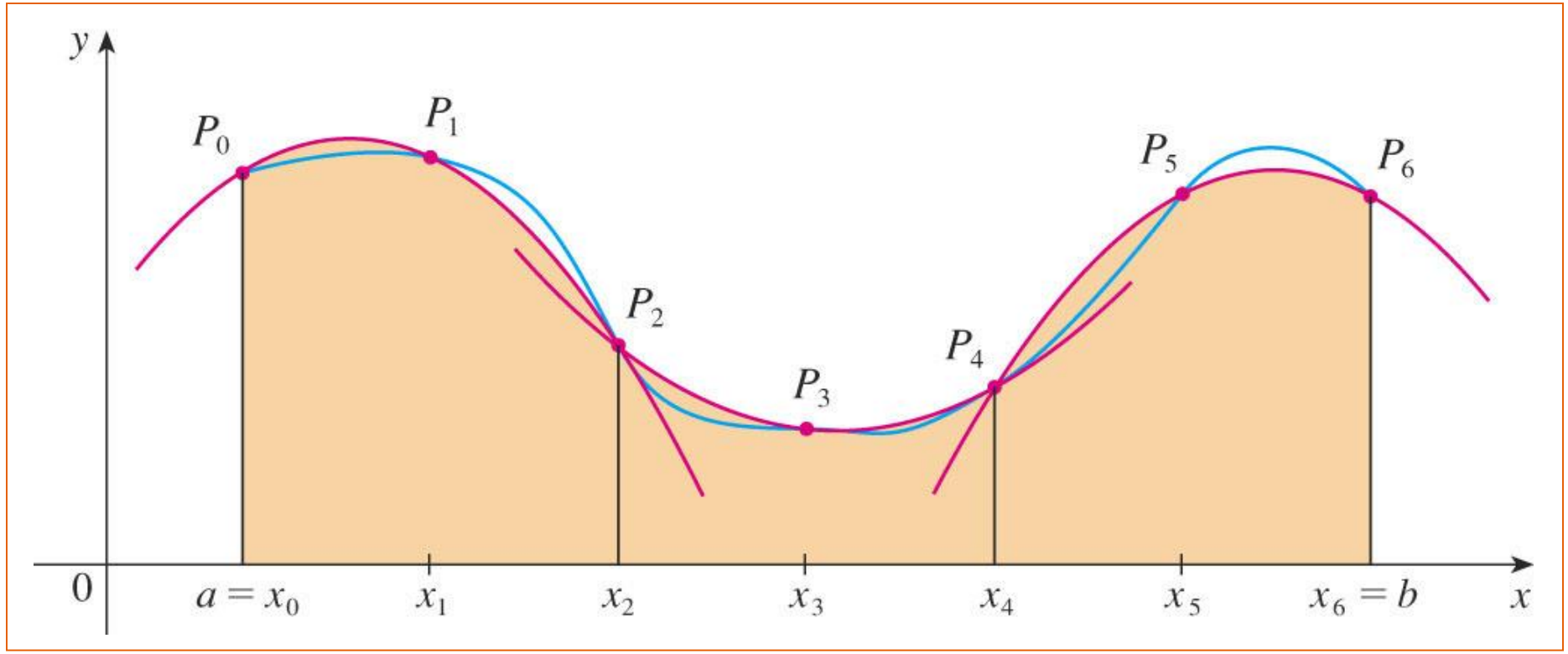
APPROXIMATE INTEGRATION

- Then, on each consecutive pair of intervals, we approximate the curve $y = f(x) \geq 0$ by a parabola, as shown.



APPROXIMATE INTEGRATION

- If $y_i = f(x_i)$, then $P_i(x_i, y_i)$ is the point on the curve lying above x_i.
  - A typical parabola/curve passes through three consecutive points: P_{i-1}, P_i, P_{i+1}



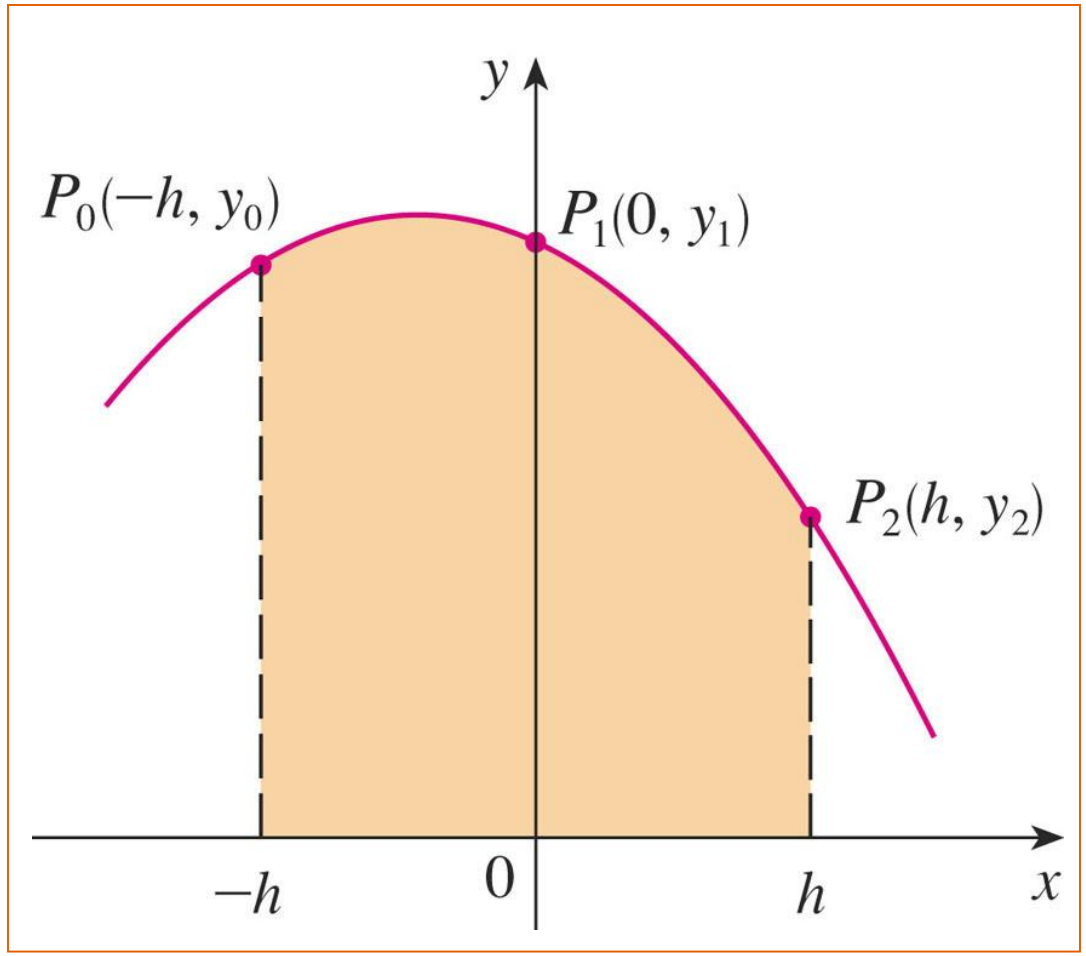
APPROXIMATE INTEGRATION

- To simplify our calculations, we first consider the case where:

$$x_0 = -h,\ x_1 = 0,\ x_2 = h$$

- We know that the equation of the parabola through P_0, P_1 and P_2 is of the form

- $y = f(x) = Ax^2 + Bx + C$



APPROXIMATE INTEGRATION

Therefore, the area under the parabola from $x = -h$ to $x = h$ is:

$$\int_{-h}^{h}(Ax^2 + Bx + C)dx = \left[A\frac{x^3}{3} + B\frac{x^2}{2} + Cx + K\right]_{-h}^{h}$$

$$= \left(A\frac{h^3}{3} + B\frac{h^2}{2} + Ch + K\right) - \left(-A\frac{h^3}{3} + B\frac{h^2}{2} - Ch + K\right)$$

$$= 2\left(A\frac{h^3}{3} + Ch\right)$$

$$= \frac{h}{3}(2Ah^2 + 6C)$$

APPROXIMATE INTEGRATION

However, as the parabola passes through $P_0(-h, y_0)$, $P_1(0, y_1)$, and $P_2(h, y_2)$, we have:

$$y_0 = A(-h)^2 + B(-h) + C$$

$$= Ah^2 - Bh + C$$

$$y_1 = C \text{ or, } 4y_1 = 4C$$

$$y_2 = Ah^2 + Bh + C$$

APPROXIMATE INTEGRATION

- Therefore,

$$y_0 + 4y_1 + y_2 = 2Ah^2 + 6C$$

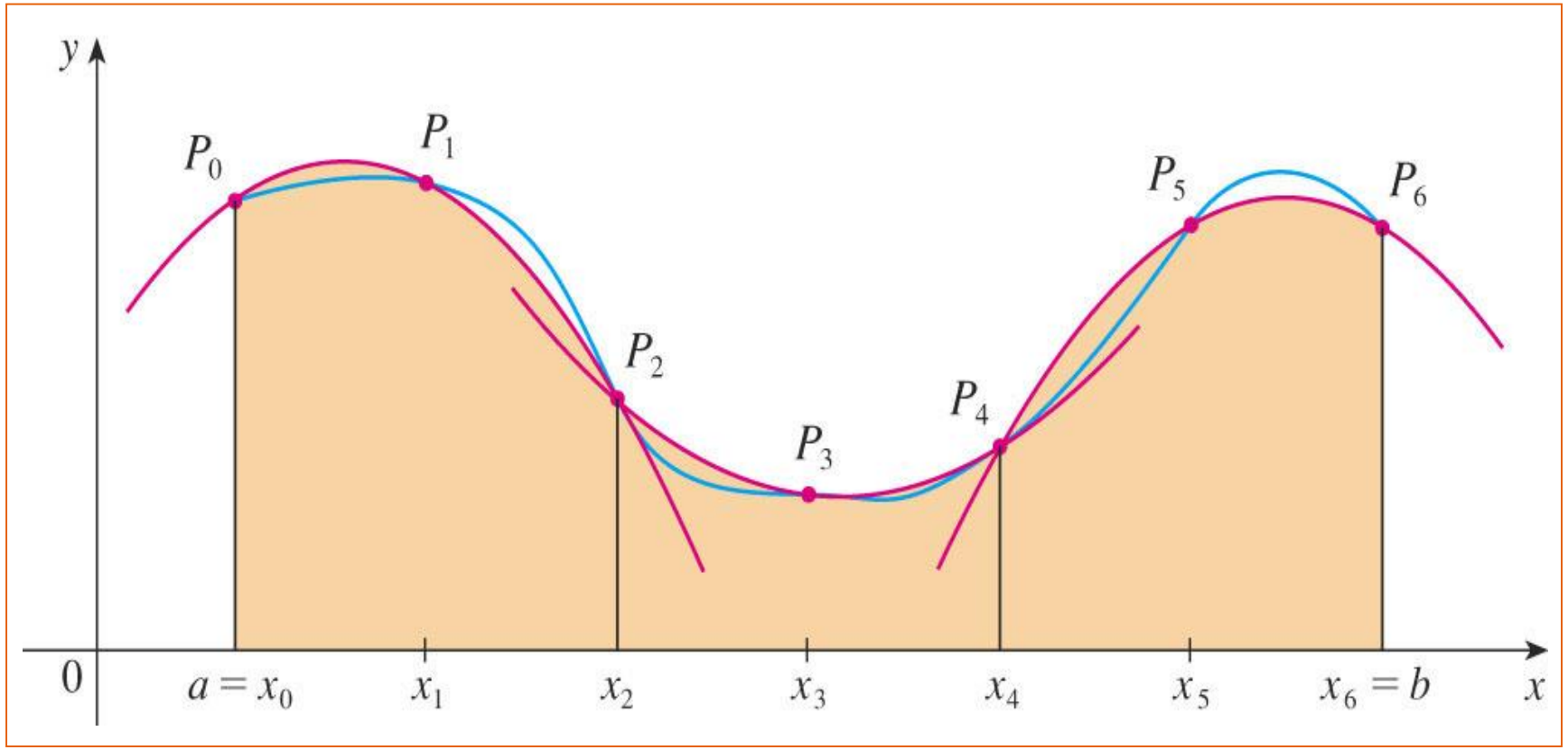
- So, we can rewrite the area under the parabola as:

$$\frac{h}{3}(y_0 + 4y_1 + y_2)$$

APPROXIMATE INTEGRATION

- Now, by shifting this parabola horizontally, we do not change the area under it.
- This means that the area under the parabola through P_0, P_1, and P_2 from $x = x_0$ to $x = x_2$ is still:

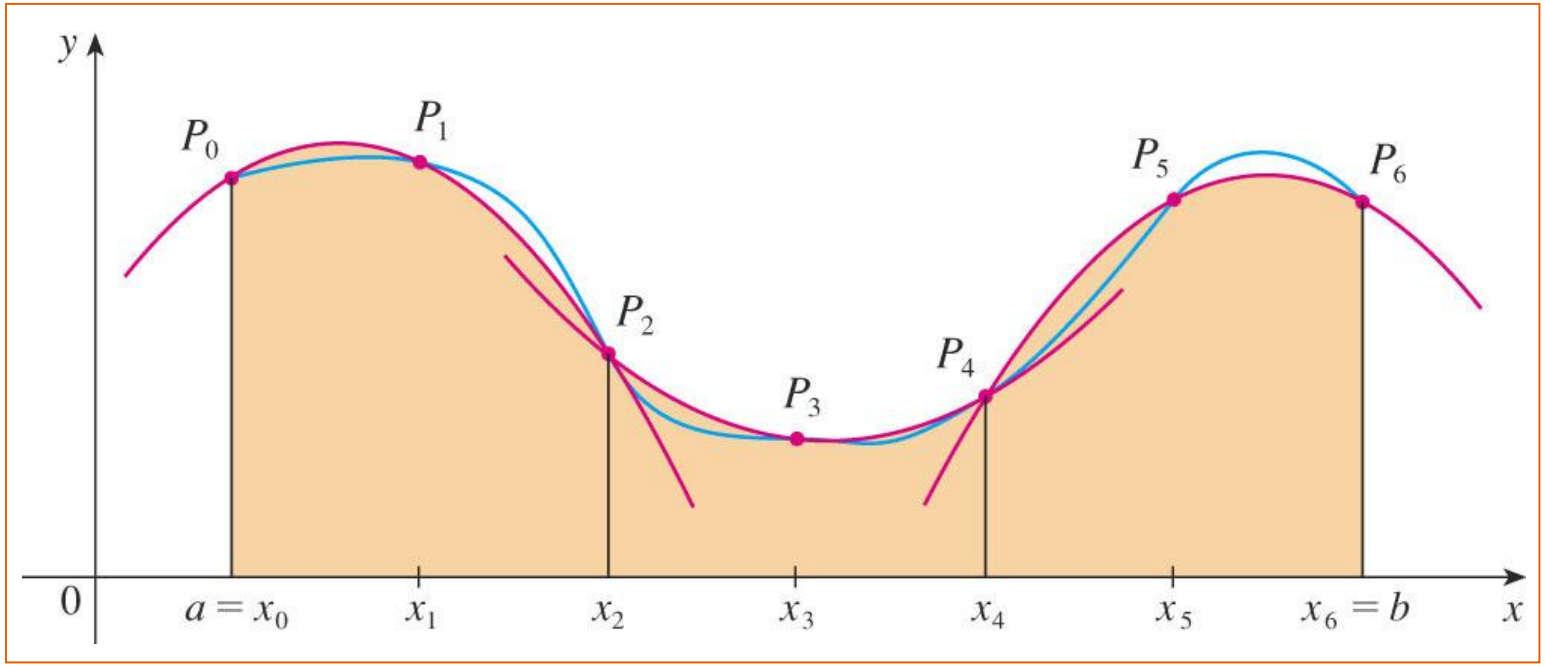
$$\frac{h}{3}(y_0 + 4y_1 + y_2)$$



APPROXIMATE INTEGRATION

- Similarly, the area under the parabola through P_2, P_3, and P_4 from $x = x_2$ to $x = x_4$ is:

$$\frac{h}{3}(y_2 + 4y_3 + y_4)$$



APPROXIMATE INTEGRATION

- Thus, if we compute the areas under all the parabolas and add the results, we get:

$$\int_a^b f(x)dx \approx \frac{h}{3}(y_0 + 4y_1 + y_2) + \frac{h}{3}(y_2 + 4y_3 + y_4) + \ldots + \frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n)$$

$$= \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + \ldots + 2y_{n-2} + 4y_{n-1} + y_n)$$

- Note the pattern of coefficients: 1, 4, 2, 4, 2, 4, 2, . . . , 4, 2, 4, 1

SIMPSON'S 1/3 RULE

- This is called Simpson's 1/3 Rule—after the English the English mathematician Thomas Simpson (1710–1761).

$$\int_a^b f(x)\,dx \approx S_n$$

$$= \frac{\Delta x}{3}[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3)$$

$$+ \ldots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$$

- where n is even and h or $\Delta x = (b - a)/n$.

SIMPSON'S RULE

Example 4

- Use Simpson's 1/3 Rule with $n = 10$ to approximate

$$\int_1^2 (1/x)\,dx$$

SIMPSON'S RULE

Example 4

- Putting $f(x) = 1/x$, $n = 10$, and $\Delta x = 0.1$ in Simpson's Rule, we obtain:

$$\int_1^2 \frac{1}{x}\,dx \approx S_{10}$$

$$= \frac{\Delta x}{3}[f(1) + 4f(1.1) + 2f(1.2) + 4f(1.3) + \ldots + 2f(1.8) + 4f(1.9) + f(2)]$$

$$= \frac{0.1}{3}\left(\frac{1}{1} + \frac{4}{1.1} + \frac{2}{1.2} + \frac{4}{1.3} + \frac{2}{1.4} + \frac{4}{1.5} + \frac{2}{1.6} + \frac{4}{1.7} + \frac{2}{1.8} + \frac{4}{1.9} + \frac{1}{2}\right)$$

$$\approx 0.693150$$

SIMPSON'S RULE

- In Example 4, notice that Simpson's Rule gives a much better approximation ($S_{10} \approx 0.693150$) to the true value of the integral ($\ln 2 \approx 0.693147$) than does either:
  - Trapezoidal Rule ($T_{10} \approx 0.693771$)
  - Midpoint Rule ($M_{10} \approx 0.692835$)

SIMPSON's 3/8 RULE

- We know that the equation of the parabola through $P_0(x_0, y_0)$, $P_1(x_1, y_1)$, $P_2(x_2, y_2)$ and $P_3(x_3, y_3)$ is of the form

$$y = f(x) = Ax^3 + Bx^2 + Cx + D$$

$$or \quad y = a_0 + a_1x + a_2x^2 + a_3x^3 \text{ (cubic polynomial)}$$

- And the area under the parabola/curve of 4 points is

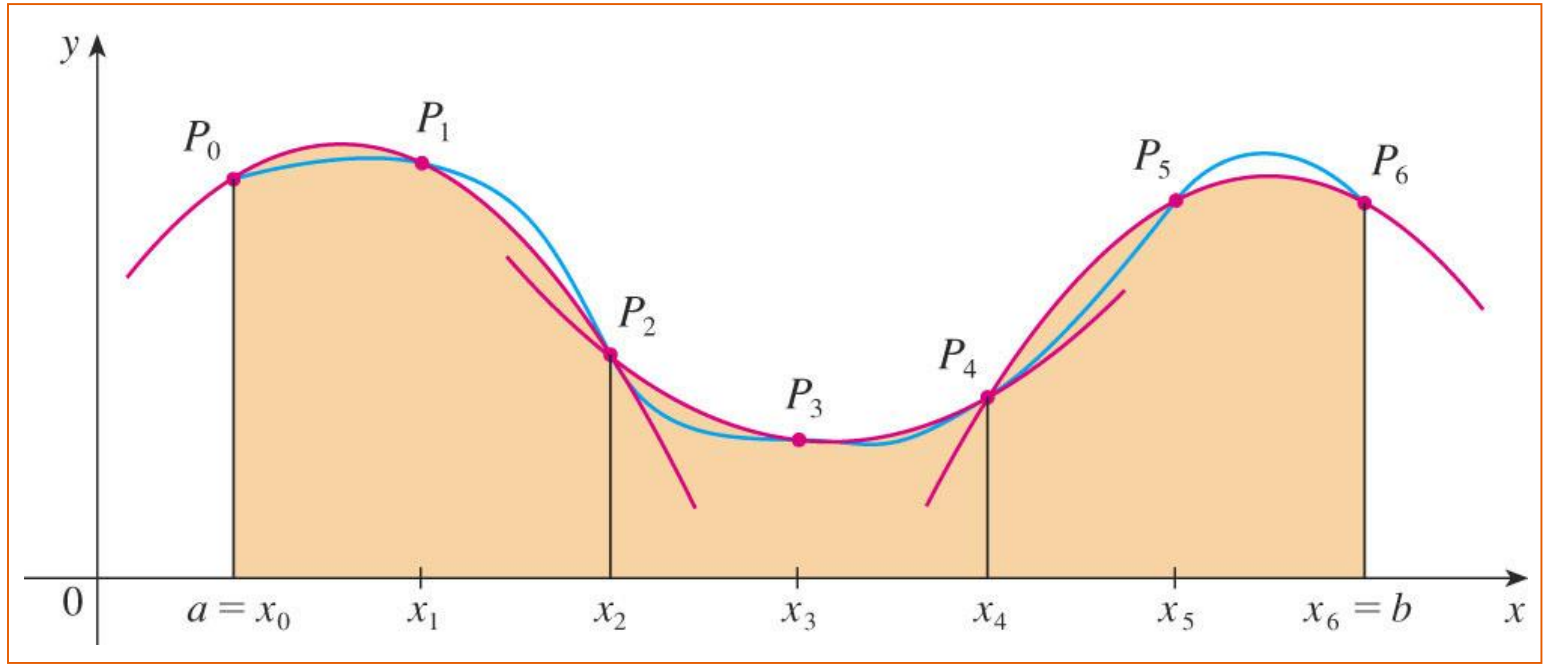
$$\frac{3h}{8}(y_0 + 3y_1 + 3y_2 + y_3)$$

SIMPSON's 3/8 RULE

- Thus, if we compute the areas under all the parabolas and add the results, we get:

$$\int_a^b f(x)dx \approx \frac{3h}{8}(y_0 + 3y_1 + 3y_2 + y_3) + \frac{3h}{8}(y_3 + 3y_4 + 3y_5 + y_6) + \ldots + \frac{3h}{8}(y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n)$$

- Note the pattern of coefficients: 1, 3, 3, 2, 3, 3, 2, . . . , 2, 3, 3, 1



SIMPSON's 3/8 RULE

- Use Simpson's 3/8 Rule with n = 6 to approximate

$$\int_1^2 (1/x)\,dx$$

- Here $\Delta x = (b-a)/n = 1/6 = 0.167$
- We have the following tables of values.

x	1	1.166667	1.333333	1.5	1.666667	1.833333	2
f(x)	1	0.857143	0.75	0.666667	0.6	0.545455	0.5

SIMPSON's 3/8 RULE

- Now, we compute the value of the integral.

$$\int_1^2 \frac{1}{x}dx \approx S_6 = \frac{3\Delta x}{8}[f(1.0) + f(2.0) + 2f(1.5) + 3\{f(1.167) + f(1.33) + f(1.667) + f(1.833)\}]$$

$$= \frac{3 \times 0.16667}{8}[1.0 + 0.5 + 2 \times 0.6667 + 3\{0.8571 + 0.75 + 0.6 + 0.5454\}]$$

= 0.06249 * 11.0911

= 0.6932